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MPJ LAB 03

Problem Statement:- Write a java program to implement method overloading and method overriding in Java.

1. Write a class Shapes. Find the area of basic shapes using polymorphism
2. WAP to create a parent class Hillstations and three subclasses by hill station names. Subclasses extend the superclass and override methods famousfood() and famousfor(). Call the methods by Hillstations class. It should refer to base class object and base class method overrides superclass method.

Objective:-

1. To study method overloading
2. To study method overriding in Java

Theory:-

1. Method overloading and its advantages

→ It allows a class to have multiple methods with the same name but different parameters, enabling polymorphism

→ Ex- int myMethod (int x)
float myMethod (float x)
double myMethod (double x, double y)

Advantages

1) Increases readability and improves maintainability
2) Offers flexibility
3) Key realization of polymorphism in Java

2. Method overriding and its rules

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→ When a subclass provides a specific implementation for a method that is already defined in its parent class.

Rules

1. Name, parameters, and return type must match parent method
2. Java picks which method to run at run time based on the actual object type, not just reference variable type
3. Static methods cannot be overridden
4. @Override annotation catches mistakes like typos in method names

Conclusion:- Thus, we have successfully implemented method overloading and method overriding in Java.

FAQs:

Q1.) Can we overload java main() method?

→ Yes, we can but JVM calls only the original main (String[] args).

Q2.) Can we declare overloaded methods as final?

→ Yes, overloaded methods can be declared as final.

Q3.) Can overloaded method be overridden?

→ Yes, a method that is overloaded in a class can also be overridden in a subclass.

Q4.) What is method overridding?

→ Method overridding in Java is used to achieve runtime polymorphism and dynamic behavior. It allows a subclass to provide its own specific implementation of a method that is already defined in its parent class.

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Q5.) What happens if we change the arguments of overriding method?

→ If we change the arguments of overidding method, it will not be overidding, it will become method overloading.

Q6.) Can we change the return type of overridding method from number type to integer type?

→ Yes, if it is a covariant return type (child type of orignal return type)